



Study Smarter API

Groq tutor backend · REST · OpenAPI 3

Mobile app is endpoint par requests bhejti hai. Firebase ads + Remote Config rakhta hai. **x-api-key** har `/api/v1/*` route par zaroori hai. Language header nahi — body/form mein `languageCode`.

3 tutor routes

11 languages

2s rate limit

Groq server-only

Production base URL

<https://studysmarter.habitrackerapi.com>

Interactive docs: `/docs` · OpenAPI JSON: `GET /api/openapi`

Authentication

HEADER

VALUE

x-api-key

Client key — server env `X_API_KEY` = Firebase `tutor_api_key`

```
curl -X POST "https://studysmarter.habitrackerapi.com/api/v1/tutor/ask" \
  -H "Content-Type: application/json" \
  -H "x-api-key: YOUR_CLIENT_KEY" \
  -d '{"prompt":"What is 2+2?","languageCode":"en"}'
```

App method → API mapping

FLUTTER (LEGACY CLIENT)

HTTP

RESPONSE FIELD

`askOpenAI(prompt)`

POST

`/api/v1/tutor/ask`

`data.text`

`analyzeImage(jpeg)`

POST

`/api/v1/tutor/image`

`data` (math or non-math JSON)

`analyzeFile(pdf)`

POST

`/api/v1/tutor/file`

`data.answer`

Firebase Remote Config

KEY	VALUE
tutor_api_base_url	https://studysmarter.habittrackerapi.com
tutor_api_key	Same as server X_API_KEY

Endpoints

POST /api/v1/tutor/ask **Text question**

Plain-text tutor answer; math with steps + Summary label. JSON body.

REQUEST

```
{
  "prompt": "Explain photosynthesis briefly",
  "languageCode": "en"
}
```

RESPONSE (200)

```
{
  "success": true,
  "message": "Answer generated.",
  "statusCode": 200,
  "data": { "text": "..." }
}
```

POST /api/v1/tutor/image **Image question**

multipart/form-data · JPEG only · max 10MB

REQUEST

```
Content-Type: multipart/form-data

image: (JPEG file)
languageCode: en
```

RESPONSE (200) — MATH

```
{
  "success": true,
  "data": {
    "answer": "...",
    "verification": "...",
    "steps": [{ "stepNumber": 1, "stepTitle": "...", "stepDescription": "..." }],
    "why": { "exampleTitle": "...", "explanation": "...", "keyPoints": [], "conclusion": "..." }
  }
}
```

RESPONSE (200) — NON-MATH

```
{ "success": true, "data": { "answer": "...", "explanation": "..." } }
```

POST /api/v1/tutor/file **PDF document**

multipart/form-data · PDF max 15MB · text extraction only (no OCR)

REQUEST

Content-Type: multipart/form-data

file: application/pdf
languageCode: en

RESPONSE (200)

```
{  
  "success": true,  
  "data": {  
    "answer": "Summary + key points + answers (one string)"  
  }  
}
```

GET /api/health **Health**

No x-api-key. Uptime monitoring.

RESPONSE (200)

```
{  
  "success": true,  
  "data": { "service": "study-smarter-api", "status": "ok" }  
}
```

App languages

Invalid or missing LanguageCode → server uses en.

en ar zh de fr pt es it tr id ms

HTTP status codes

CODE	MEANING	WHEN
200	Success	Standard envelope with data
400	Bad request	Validation, wrong file type, unreadable PDF
401	Unauthorized	Missing or invalid x-api-key
405	Method not allowed	Tutor routes are POST-only
429	Rate limited	Min 2 seconds between requests per key
502	Upstream / format	Groq error or AI output validation failed